

$$\sum_{l=1}^L \|J_l\|_F^2$$

|| *w* || 2

Unconstrained Jacobians

$$\min_{w, \{J_l\}_{l=1}^L} \mathcal{L} \left(w^T \prod_{l=1}^L (I + J_l) X, y \right) + \frac{\lambda}{2} \sum_{l=1}^L \|J_l\|_F^2 + \frac{\lambda}{2} \|w\|_2^2$$

$$\|w\|_2^2$$

Typical weight decay on classifier

$$\sum_{l=1}^L \|J_l\|_F^2$$

J_i are all rank-one

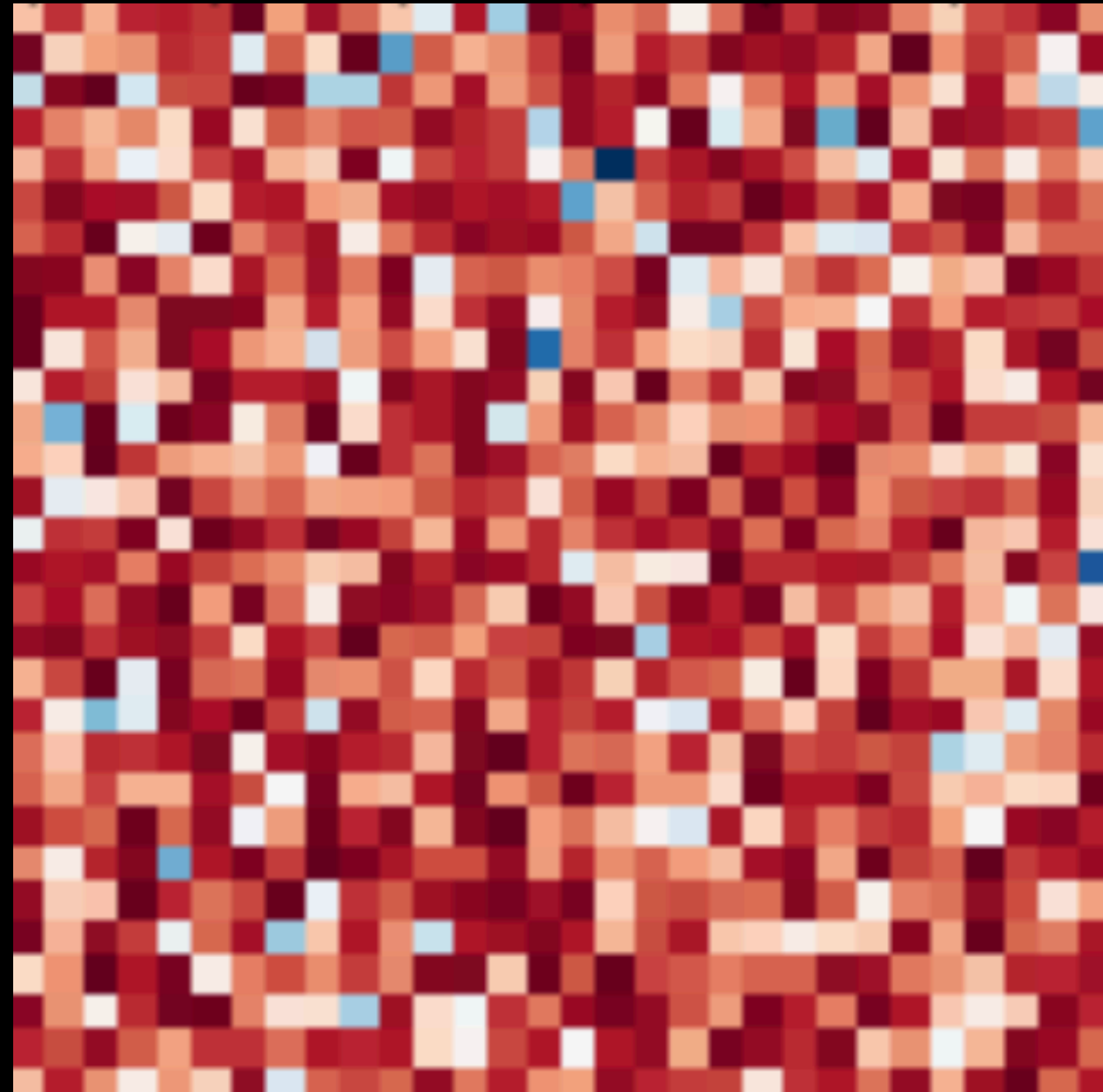
Top singular vectors of J_i are aligned

Top singular values of J_i are equal

Trajectories are perfectly linear

Token-wise Coupling: Visualization of A

Llama-3 8B (Untrained)



Llama-3 8B

